



PORTLAND POLICE BUREAU UNMANNED AERIAL SYSTEMS (UAS) Annual Report 2025

BACKGROUND

The Portland Police Bureau has continued its work to field and implement a structured, transparent, and safe Unmanned Aerial Systems (UAS) program to enhance service delivery and improve safety within the City of Portland.

The program began to take shape in 2021 through planning efforts led by the Metropolitan Explosives Disposal Unit (MEDU) and Air Support Unit (ASU). Following community engagement, development of a Standard Operating Procedure (SOP), and completion of a Privacy Impact Assessment (PIA), Portland City Council authorized a one-year pilot project in April 2023 through Ordinance 191224. The pilot concluded in April 2024 and demonstrated measurable benefits, including improved situational awareness during critical incidents and significant reductions in time spent documenting major crash scenes.

After reviewing the pilot findings, City Council approved expansion of UAS operations to the Bureau's three precincts through Ordinance 191882. Precinct-based deployment began November 1, 2024.

Calendar year 2025 represents the first full year of sustained precinct-based operations. The program has transitioned from a limited pilot initiative into a regularly utilized public safety tool supporting patrol response, traffic investigations, tactical operations, and specialized unit functions. This transition did not expand the scope of authorized uses; rather, it reflected the availability of trained operators across precincts and shifts, allowing UAS to be deployed more consistently within established policy guidelines.

In late September 2025, the Bureau launched a limited Drone First Responder (DFR) pilot program in partnership with the Gresham Police Department. Operating from a fixed site at East Precinct, the DFR model allows for immediate drone deployment upon dispatch to qualifying calls for service, providing aerial situational awareness prior to officer arrival. An additional Privacy Impact Assessment was completed to support the pilot, which is being evaluated to determine operational effectiveness, response efficiency, and future feasibility.

TRAINING

Following expansion of the program in 2024, precinct-based remote pilots obtained their FAA Part 107 certifications and completed structured operator training focused on safety, regulatory compliance, and tactical flight concepts. During calendar year 2025, the Bureau

trained six additional pilots to support operational needs and ensure consistent coverage across precincts and shifts.

All UAS operators within the Portland Police Bureau maintain current FAA Part 107 certification, which requires demonstrated knowledge of airspace regulations and successful completion of recurrent aeronautical knowledge testing every 24 months. In addition to federal requirements, each pilot is subject to internal flight hour standards, regular skills evaluations, and structured maintenance training.

Throughout 2025, the Bureau conducted monthly UAS training sessions. Each pilot attended training every other month, resulting in six full-day training sessions per pilot annually. Training emphasized safe flight operations, tactical integration with patrol and specialized units, coordination during high-risk incidents, DFR procedures, and reinforcement of privacy and policy standards.

To remain aligned with best practices and emerging regulatory developments, the program supervisor attended the 2025 Law Enforcement Drone Association (LEDA) conference. Attendance provided exposure to peer-agency operations, evolving technologies, and national standards related to public safety drone programs.

DEPLOYMENTS: AUTHORIZED USES / PROHIBITIONS

UAS deployments continue to be governed by the Bureau's Standard Operating Procedure (SOP), Police Bureau Directives, and applicable federal and state laws and regulations. Following the conclusion of the initial pilot project and subsequent City Council approval, the current SOP reflects authorized deployment criteria consistent with Oregon law and established privacy safeguards.

Current, **allowed** uses for UAS technology continue to include:

- Pursuant to a valid warrant authorizing its use
- Where there is probable cause or belief that a person has committed a crime, is committing a crime, or about to commit a crime, and exigent circumstances exist that make it unreasonable to obtain a warrant authorizing its use
- With written consent of an individual for the purpose of acquiring information about the individual or the individual's property
- As part of search and rescue activities

- For the purpose of assisting an individual in an emergency where there is a reasonable belief there is an imminent threat to the life or safety of the individual
- During a state of emergency declared by the Governor
- For the purpose of reconstruction of a specific crime scene, or accident scene, or a similar physical assessment, related to a specific investigation
- For training in the use and acquisition of information

Prohibited uses and privacy considerations include:

- Conducting random or indiscriminate mass surveillance activities
- Targeting a subject based on race, ethnicity, national origin, religion, disability, economic source or status, housing status, gender, or sexual orientation
- Harassment, intimidation, or discrimination against any individual or group
- Personal business
- Crowd control/crowd management unless a life safety critical incident occurs
- Weaponization of any UAS
- Any use of facial recognition technology

UAS CAPABILITIES / PROCUREMENT / INVENTORY

At the conclusion of calendar year 2025, the Police Bureau had an inventory of thirty-six drones. Twenty-four of these are small drones, which are inexpensive and available for public/consumer purchase and use. These drones are generally used as trainers and for interior flights. Four drones are assigned to crash investigation and reconstruction within the Traffic Division. The inventory currently includes ten drones that can fly during inclement weather and in the dark. The Police Bureau currently has 44 trained and certified UAS pilots.

Primary equipment used by PPB: PPB UAS operates the *DJI M4TD* as its primary overwatch and tactical drone. This drone is (Ingress Protection) IP rated for rain and can operate in winds up to 40mph. It has color and infrared cameras, obstacle avoidance and an excellent control unit. With the batteries and tools needed, the M4TD costs roughly \$15,000. PPB currently has four M4TD UAS.

PPB also operates six *DJI Matrice 30T* UAS, which have similar capabilities as the M4TD, but are an older platform. The M4TD cost roughly \$12,000 per unit.

For interior flight, training, and short flights in the daylight, the *DJI Avata*, *DJI Mini 2*, *DJI Neo*, and *DJI Flip* are utilized. These drones are all consumer-oriented and cost under \$1000 each.

The Police Bureau began efforts to upfit six marked police vehicles for precinct-based UAS operations. The Police Bureau did not take delivery of the purpose-designed vehicles in 2025, but anticipates delivery in early 2026.

TRANSPARENCY

Police Bureau Website

The Police Bureau website maintains a page about the UAS program (<https://www.portland.gov/police/community/drones>). This site covers the purposes and background of the UAS program, along with links to the SOP and City Council session videos.

Public-Facing Data Dashboard

A public dashboard has been in existence since the inception of the UAS program. The dashboard contains information on all UAS deployments, including call type, location and other metrics (<https://www.portland.gov/police/open-data/uascalls>). Data points displayed within the page are updated on the fifteenth of each month and the page has been viewed over 6,700 times.

FLIGHTS / UTILIZATION

In calendar year 2025, the program deployed UAS technology on 774 occasions, representing the first full year of precinct-based operational integration. Deployments averaged approximately 65 missions per month throughout the year, reflecting normalized availability of trained operators across all three precincts. Of the 2025 deployments, 516 flights (67%) were categorized as exigent circumstances, meaning requests from patrol or tactical teams for assistance during high-risk or rapidly evolving incidents. Training flights accounted for 155 deployments (20%), underscoring continued emphasis on proficiency and operator development during program growth. Search warrant support comprised 40 deployments (5%), while traffic-related deployments accounted for 31 missions (4%). The remaining use cases included Signed Consent (23 deployments, 3%), Search and Rescue (6 deployments, less than 1%), and Explosive-related incidents (3 deployments, less than 1%).



Unmanned Aerial Systems Call Activity

Strategic Services Division | Portland Police Bureau



Data refreshed on the first business day after the 15th of the month

Drones per Call

1.5

Call Group

- Alarm
- Civil
- Community Policing
- Crime
- Disorder
- Other
- Traffic

Report Month & Year

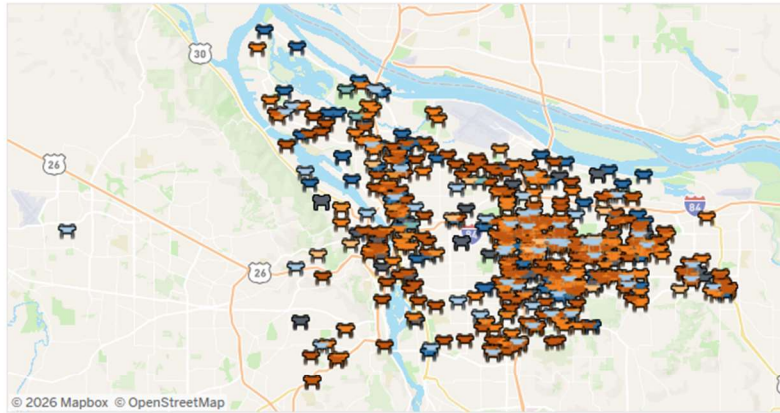
(Multiple values)



Calls by Category

Note: flights not associated with a call will not have a mapped location

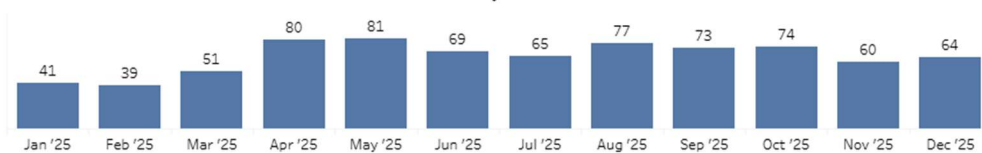
Grand Total	774
Suspicious	774
Detail Patrol	774
Disturbance	774
Other	774
Traffic Stop	774
Warrant	774
Collision	774
Burglary	774
Follow Up	774
Assault	774
Non-Call Flights (e.g. Training)	774
Escape/Pursuit	774
Theft	774
Area/Premise Check	774
Assist	774
Stolen Vehicle	774
Unwanted Person	774
Shots Fired	774
Threat	774
Welfare Check	774
Person Contact	774
Robbery	774
Hazard	774
Community Policing	774
Street Racing	774
Alarm	774
Behavioral Health	774
Admin	774
Restraining Order Violation	774
TrillMet	774
Vandalism	774
Arson	774
Missing Person	774
Vice	774
Disorder	774
Domestic Violence	774
Juvenile Problem	774
School Incident	774
Abuse	774
Civil	774
DUI	774
Parking Problem	774
Subject Stop	774



Calls by Flight Type

Exigent	516
UAS Training	155
Search Warrant	40
Traffic	31
Signed Consent	23
Search & Rescue	6
Explosive Device	3

Calls by Month



HIGHLIGHTS

The Police Bureau continues to see benefits from the UAS program, to include reduced time spent at crash and critical incident scenes, improved de-escalation of armed subjects, and enhanced situational awareness, all contributing to better overall outcomes. The Bureau has found many successful ways to enhance community safety with UAS technologies.

- Traffic Investigations:** The Traffic Division continues to see significant improvements in reduced time spent documenting major crash scenes, and the quality and accuracy of evidence gathered. The pilot program noted an approximately 80% decrease in time spent mapping scenes. This allows police to clear crash scenes more quickly, which significantly lessens overall impacts to the traveling public. Additionally, traffic investigators have collected “fly-thru” videos of vehicle trajectories and overhead views of crash locations, which have been exceptionally helpful when presenting cases to Grand Jury or in courtroom trials. PPB Traffic Division conducted 31 evidence collection flights in 2025, supporting investigators from PPB, the Amtrak Police Department, and the Multnomah County Sheriff’s Office.

- *Explosive related incidents:* Members of our regional Explosive Disposal Unit continue to utilize UAS to safely gather information on suspicious items and confirm the area is clear of potential victims. In February and March 2025, EDU technicians utilized UAS as part of operations to dispose of unexploded military ordnance located during a construction project in Vancouver, Washington. In April 2025, an EDU technician utilized UAS to record the safe destruction of an explosive device in a natural area. Video captured by the UAS led to recovery of evidence from a creek bed.
- *Patrol:* Patrol Officers deploy UAS to crimes in progress and exigent circumstances. UAS technology provides officers with additional time and distance from potential threats, which provides additional options to officers as they navigate high-risk encounters with armed suspects. A notable deployment highlights this benefit:
 - August 23, 2025 – UAS Officers responded to back-to-back calls requiring their service. The first call was a theft from a residential property, where the suspect fled on foot when approached by the first officers on scene. UAS pilots located the suspect on a rooftop within minutes of beginning their search, and the suspect was arrested shortly thereafter. While this call was concluding, the Bureau of Emergency Communications (BOEC, Portland's 911 center) received a report of an armed individual firing a handgun inside a family member's home. The subject went outside and another community member called to report the subject was still armed and entering the community member's property. When officers arrived in the area, they utilized UAS to locate the subject, who began breaking the side door of an occupied neighbor's home. UAS pilots observed the subject conceal himself in a garbage bin and were able to coordinate a deliberate and planned response to the incident. The subject ultimately surrendered to officers without any use of force.



- *Tactical Incidents:* UAS technology continues to be utilized to provide exterior overwatch and assist with interior searches for the Special Emergency Reaction Team (SERT) and Crisis Negotiation Team (CNT) during high-risk search warrants and full-team activations. UAS offers real-time information on what lies ahead for members searching for armed suspects indoors and outdoors. This situational knowledge provides critical information for tactical decision making which can greatly reduce the need for force. The use of UAS also routinely reduces time spent on these events which can be disruptive and unsettling for community members.
- *Drone First Responder (DFR) Pilot Rollout:* Building upon the operational success of precinct-based integration, the Bureau launched a Drone First Responder (DFR) pilot program in late September 2025 in partnership with the Gresham Police Department. The DFR model could prove to be a forward-thinking evolution of the program's capabilities.

Under the DFR pilot framework, a drone is pre-positioned at a fixed launch site at East Precinct and may be deployed immediately upon dispatch to qualifying calls for service. The objective is to provide supervisors and responding officers with near-real-time aerial situational awareness prior to arrival on scene. The pilot is designed to assess whether rapid aerial deployment can:

- Improve resource allocation decisions
- Enhance officer safety during initial approach
- Reduce uncertainty in high-risk or rapidly evolving incidents
- Potentially eliminate the need for certain ground responses when aerial assessment provides sufficient information

Recognizing the expanded scope associated with the DFR program, the Bureau partnered with Smart City PDX for an updated Privacy Impact Assessment specific to DFR operations. This updated review examined data collection practices, flight boundaries, operational safeguards, and oversight mechanisms unique to first-responder deployment models. The DFR pilot is being evaluated through data collection focused on response times, deployment frequency, and call outcomes.

The pilot program remains geographically limited to a single operational site, allowing the Bureau to measure outcomes and refine procedures before considering broader implementation.

Early metrics from the PPB DFR pilot indicate that DFR drones arrive on scene approximately five times faster than patrol officers to the same calls. DFR drone pilots resolve nearly one in five calls without requiring a patrol officer to arrive on scene, freeing responders to head to higher priority incidents where they are urgently needed.

Travel Time (min) Dispatch to At-Scene		DFR Clears 27 (18.5%)
DFR	2.8	
Other Unit	16.9	



LESSONS LEARNED

There was significant progress in our agency's integration of Unmanned Aircraft Systems (UAS) technology during 2025. Through numerous deployments and ongoing evaluation, we have identified areas for ongoing improvement and growth, ultimately enhancing our ability to serve our community and increase safety.

- *Transformation from new tool to essential resource:*

Officers faced with searches for fleeing suspects and high-risk incidents have grown to expect timely and accurate information provided by UAS. Since UAS pilots are assigned to all shifts and precincts, this service has quickly become a core part of the resolution of these incidents. As a result, the UAS program saw a 265% increase in missions flown from 2024 (212 missions) to 2025 (774 missions). However, when at work, UAS pilots are primarily responsible for staffing a patrol district. Their

assignment as a UAS pilot is an additional duty, and their availability to fly is subject to their primary responsibilities being completed. The UAS program is evaluating options to increase UAS availability and avoid overloading short-staffed patrol officers.

- *Optimizing UAS Operator Deployment and Equipment:*

The effectiveness of precinct-based UAS operators is directly tied to their ability to rapidly deploy with all necessary equipment. Standard patrol vehicles are not optimally configured for the efficient transport and deployment of UAS equipment. We are prioritizing the outfitting/retrofitting of vehicles for dedicated UAS operators. These vehicles will be designed to allow for rapid deployment, secure equipment storage, and on-site operations.

This project is ongoing, and the UAS program has identified the need for more robust data connectivity in its UAS vehicles.

- *Integration into Crime Reduction Initiatives:*

UAS technology provides a powerful tool for enhancing crime reduction efforts through mission-based operations. Integrating UAS technology into missions to address stolen vehicles [Stolen Vehicle Operations (SVO)] and retail theft has proven effective.

- *Training and Standardized Operating Procedures (SOPs):*

As UAS operations expand, consistent training and standardized operating procedures continue to be crucial for ensuring safe, effective, and lawful deployments. Ongoing training will remain essential to maintain operator proficiency, and clear SOPs provide guidance for the many considerations that come with operating UAS equipment.

The lessons learned this year will serve as a foundation for continued advancement in our UAS program. We are committed to leveraging this technology responsibly and effectively to enhance public safety.

RECOMMENDATIONS

Based on the insights gained during the past year, the following actions are recommended to further enhance our UAS program and increase its positive impact on public safety:

- *Planning for Future Operations and Federal Considerations:*

The Bureau needs to continue planning for long-term program sustainability. An important component of that planning involves navigating federal regulatory changes concerning foreign-manufactured unmanned aircraft systems.

The Bureau's current fleet consists entirely of DJI-manufactured aircraft. In December 2025, the Federal Communications Commission (FCC) expanded its Covered List to include foreign-manufactured unmanned aircraft systems, effectively preventing new FCC equipment authorizations for foreign-built drone models. While the ruling does not prohibit continued operation of aircraft already authorized and in service, it restricts the approval and importation of future foreign-manufactured platforms. This regulatory shift may significantly affect the Bureau's ability to purchase replacement aircraft or expand its existing fleet in the coming years.

Although no operational changes were required during 2025, Bureau leadership initiated preliminary planning efforts to assess potential impacts. These considerations include:

- Evaluating domestically manufactured UAS platforms
- Assessing interoperability with existing evidence management systems
- Estimating financial implications of fleet replacement
- Developing contingency procurement strategies

The Bureau will continue to monitor federal regulatory developments and assess how any changes may affect future UAS procurement and fleet sustainability. As the program continues to evolve, including the implementation and evaluation of the DFR pilot, the Bureau remains committed to balancing operational capability with careful planning to ensure long-term continuity and compliance.

- *Continued Research and Development:*

Continue to evaluate emerging UAS technologies and operational applications to ensure the program remains current, effective, and aligned with evolving public safety practices and regulatory requirements.

- *Drone as a First Responder (DFR) Operations:*

Continue evaluation of the Drone First Responder (DFR) pilot program to determine its operational effectiveness and overall impact on patrol response. Ongoing data collection should focus on response times, deployment frequency, officer safety considerations, and call outcomes to ensure the program meets its intended

objectives. Findings from the pilot should be documented and used to inform future decisions.

The Bureau should also assess what would be required to implement a DFR model capable of supporting broader geographic coverage. This review should consider potential launch site locations, facility and infrastructure needs, staffing and training requirements, equipment sustainability, and associated costs. Any future expansion should remain consistent with established SOP guidelines, Police Bureau Directives, applicable laws and regulations, and existing privacy safeguards to ensure responsible and sustainable implementation.